

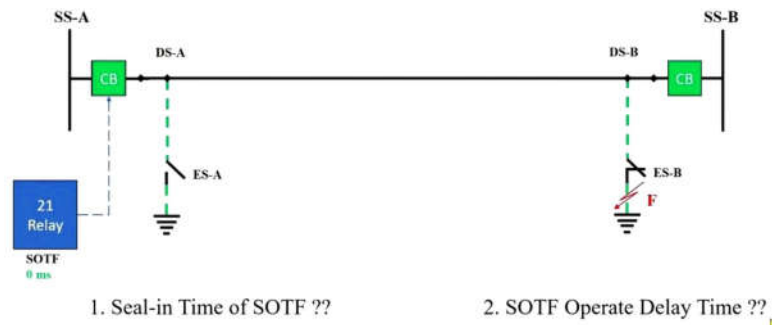
Switch On To Fault Protection

Basic Concepts:

SOTF (Switch On to Fault) is a protection function designed to prevent a circuit breaker from being closed onto an existing fault on the power line or equipment.

When the circuit breaker is closed (manually or by remote command) while a fault already exists on the system, the SOTF function detects it instantly and issues a trip or block signal, protecting the system from severe damage.

Switch On To Fault – SOTF (Practically)



Purpose of SOTF Protection:

- To avoid energizing a faulted line during breaker closing.
- To protect equipment (CT, PT, transmission line, transformer, etc.) from high fault currents.
- To maintain system stability by preventing unnecessary disturbances.
- To ensure safety for personnel and system components.

Operating Principle:

1. When a breaker close command is issued (manually or automatically).
2. The relay instantly checks for any fault condition (voltage, current, impedance).
3. If fault is present → relay issues trip/block signal (CB will not close or will open immediately).
4. If no fault is present → CB closes normally.

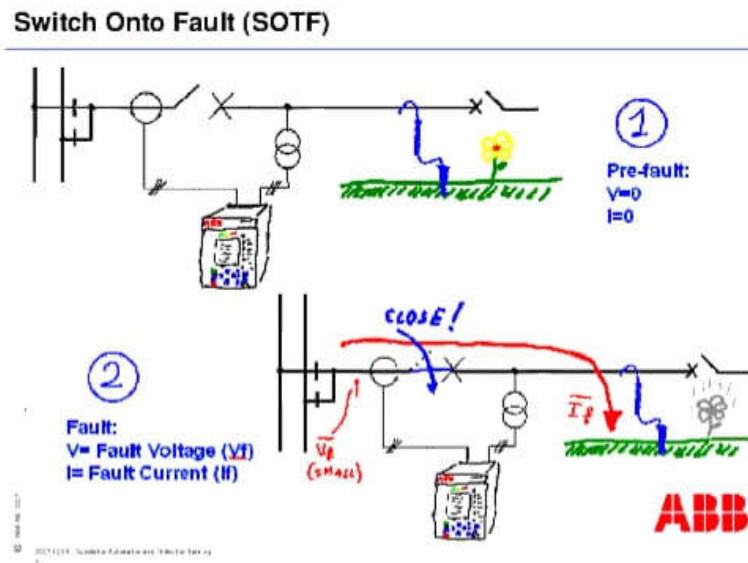
Real-Time Example:

Scenario:

A transmission line from Bus A to Bus B is under maintenance. Fault occurs at some point on the line (for example, phase-to-ground). After maintenance, the operator tries to manually close the CB at Bus A. SOTF function inside the relay detects low impedance / fault condition at the line end. Within 34.4 milliseconds, the relay sends a trip or block signal.

Result: CB does not energize the faulted line → System remains safe.

- Observation:
- Trip time = 34.4 ms
- Lockout = OK
- SOTF worked perfectly.



SOTF Testing Procedure:

1. Simulate a fault (e.g. phase-to-ground) while CB is open, Fault present
2. Give a manual close command to CB, Relay detects fault instantly
3. Observe relay operation CB trip/block within few milliseconds
4. Clear fault and close CB again CB closes normally
5. Record results Trip time, Lockout, Status OK

Typical Test Report Example:

Parameter	Observation	Result
Test Type	21 SOTF by Manual Close	
Trip Time	34.4 ms	✓ Within limit
Lockout/CB Trip	OK	✓ SOTF Active
Remark	SOTF function verified successfully	✓ PASS

Important Points to Remember:

- SOTF function only operates during breaker closing (not during normal running condition).
- It is usually linked with distance or overcurrent protection (21, 50/51).
- SOTF time response is very fast (typically 20–50 ms).
- Always perform test with load disconnected for safety.
- False SOTF operation may occur if CT/VT wiring or settings are incorrect.

Difference Between Normal Trip and SOTF Trip:

Feature	Normal Protection	SOTF Protection
Fault occurs after breaker is ON	Trip by normal protection (21, 50/51, etc.)	
Fault exists before breaker is ON		Trip/block by SOTF
Operation Condition	During running	During closing attempt
Objective	Clear running fault	Prevent energizing fault